

DPSim Unified Documentation Audit And Consolidation

Audit date: 2026-04-25 **Scope:** Downstream Processing Simulator user manual, Appendix J functionalization protocols, quickstart, configuration reference, and documentation index.

Authoritative Documentation Set

The DPSim user-facing documentation set is divided by role:

Role	Authoritative file
Operator manual and lifecycle workflow	<code>polysaccharide_microsphere_simulator_first_edition.md</code>
Chemical functionalization wet-lab protocols	<code>appendix_J_functionalization_protocols.md</code>
Installation and first-run guide	<code>../quickstart.md</code>
TOML and <code>ProcessRecipe</code> parameter reference	<code>../configuration.md</code>
Documentation navigation	<code>../INDEX.md</code>

The current simulator contract is lifecycle-first: target product profile, M1 fabrication, M2 functionalization, M3 column method, lifecycle run, validation/evidence review, and calibration comparison. Legacy M1/L1-L4 language remains only as an explanation of inherited numerical kernels.

Corrections Applied

Area	Finding	Correction
Product identity	The manual still described a polysaccharide-only emulsification simulator.	Reframed DPSim as a lifecycle downstream-processing simulator for M1/M2/M3 affinity media.
UI workflow	The manual described isolated Module 1/2/3 tabs.	Replaced quickstart with the seven-step lifecycle workflow now implemented in Streamlit.
Evidence terminology	Documentation mixed old uppercase shorthand and current enum values.	Defined canonical values: <code>validated_quantitative</code> , <code>calibrated_local</code> , <code>semi_quantitative</code> , <code>qualitative_trend</code> , <code>unsupported</code> .
Recipe source of truth	Some docs implied direct legacy controls were authoritative.	Stated that lifecycle work is driven by <code>ProcessRecipe</code> and <code>dpsim.core.recipe_io</code> .

Area	Finding	Correction
M2 protocol mapping	Appendix J contained old dropdown names, "not wired" notes, and outdated version labels.	Replaced with implemented M2 templates and reagent_key names; external chemistries are explicitly marked non-quantitative.
Wet-lab release gates	The manual lacked a consolidated assay gate before using M1/M2/M3 outputs.	Added M1 DSD/residual/pore/mechanics, M2 ligand/site/leaching, and M3 pressure/breakthrough/recovery gates.
DCM safety	PLGA protocol understated DCM hazards.	Updated DCM handling to fume-hood, suspected carcinogen/CNS hazard, compatible gloves, halogenated waste, and no odor-based clearance.
Epichlorohydrin safety	Appendix J used odor as a wash endpoint.	Replaced with residual check or validated wash endpoint; odor is not a clearance test.
EDC/NHS operation	The manual implied PBS wash alone "quenches" after coupling.	Clarified ligand wash, then ethanolamine or Tris quench after intended coupling; azide limitations added.
Runtime paths	User-facing docs did not clearly list path overrides.	Added DPSIM_TMPDIR , DPSIM_CACHE_DIR , and DPSIM_OUTPUT_DIR .
Calibration linkage	Docs did not state that M3 evidence is capped by M2 media quality.	Added evidence-governance language to manual and Appendix J.

Canonical Terms

Use these terms consistently:

Preferred term	Avoid
Downstream Processing Simulator or DPSim	EmulSim branding in user-facing docs
functionalization	alternate British spelling in new DPSim docs
M1 fabrication	Level 1 as a user workflow
M2 functionalization chemistry	Module 2 tab-only workflow
M3 column method/performance	mechanical-only chromatography wording
ProcessRecipe	temporary bridge, form-only state, or implicit UI state
reagent key	dropdown key, old module key
evidence tier	trust badge without provenance
development SOP scaffold	GMP batch record

Scientific And Operational Boundaries

DPSim documentation must not imply that simulation alone releases a process for manufacture, clinical use, or regulated quality decisions. Every lifecycle result must be interpreted with:

- unit consistency checks;
- pH, temperature, reagent, and buffer compatibility;
- calibration-domain checks;
- M1 residual oil/surfactant carryover limits;
- M2 site and mass balance;
- M2 ligand density, leaching, free-protein wash, and activity-retention assays;
- M3 pressure, compression, Reynolds/Peclet, breakthrough, and recovery checks;
- explicit assumptions and wet-lab caveats.

Implemented M2 Templates

The documentation now treats these as implemented M2 recipe templates:

Template	Required bench confirmation
<code>epoxy_protein_a</code>	Epoxide activation or hold-time control, Protein A density, retained IgG binding, free Protein A in washes
<code>edc_nhs_protein_a</code>	Carboxyl spacer density, NHS activation freshness, Protein A density/activity, free protein
<code>hydrazide_protein_a</code>	Hydrazide density, oxidized Protein A handling, activity/leaching
<code>vinyl_sulfone_protein_a</code>	Vinyl sulfone activation, nucleophile carryover control, thiol/amine coupling activity
<code>nta_imac</code>	NTA density, nickel loading, metal leaching
<code>ida_imac</code>	IDA density, nickel loading, higher metal-leaching risk than NTA

Chemistries without an implemented **ReagentProfile** remain valid wet-lab protocols when executed by trained researchers, but DPSim must treat their simulation contribution as external evidence or **unsupported** until calibrated.

Remaining Documentation Risks

The user-manual set is now aligned to current implementation, but the broader historical **docs/** tree still contains archival planning files with version history. Those files are useful for traceability and should not be read as current operating instructions unless **docs/INDEX.md** marks them as user-facing or canonical.

Future documentation updates should be gated by:

- 1 `rg` scan for old branding, old version-specific UI claims, and "not wired" language in user-facing docs.
 - 1 Cross-check of every documented reagent key against `src/dpsim/module2_functionalization/reagent_profiles.py`.
 - 1 Cross-check of every lifecycle workflow statement against `src/dpsim/visualization/app.py` and `src/dpsim/visualization/ui_workflow.py`.
 - 1 PDF rebuild of every Markdown target in `docs/user_manual/build_pdf.py`.
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Addendum — Changes since 2026-04-25 (added 2026-05-01)

This addendum records material changes to the documentation set that occurred *after* the 2026-04-25 audit was finalised. The audit body above is preserved as a point-in-time snapshot; this section is append-only.

Scope additions to the authoritative documentation set

Role	New authoritative file	Status
M1 CFD-PBE zonal coupling — operator chapter	<code>polysaccharide_microsphere_simulator_first_edition.md</code> §9	New chapter inserted between §8 (Calibration) and Part III (Appendix)
M1 CFD-PBE zonal coupling — advanced reference	<code>appendix_K_cfd_pbe_zonal_coupling.md</code>	New sibling appendix; registered in <code>build_pdf.py</code> BUILD_TARGETS
<code>zones.json</code> schema v1.0 contract	<code>cad/cfd/zones_schema.md</code>	Locked 2026-05-01; semantic versioning policy documented

Releases since the audit

Version	Date	Highlights
v0.6.0	2026-05-01	CAD geometry source-of-truth for the 5 wetted parts (Stirrer A pitched-blade with 19-tab disk geometry verified by photo + manual review; Stirrer B rotor + stator with 72 perforations across 24 columns × 3 rows — column count revised 12 → 24 on 2026-05-04; 100 mm beaker; jacketed 92 mm vessel). OpenFOAM CFD-PBE pipeline scaffolding (<code>cad/cfd/</code> , <code>src/dpsim/cfd/</code>). Live Stirrer A cross-section animation with physically faithful flow / droplet behaviour
post-v0.6.0 (commit a5d984c, 2026-05-01)	2026-05-01	DPSim-side CFD-PBE zonal coupling: schema v1.0, Pydantic-validated loader (11 hard-validation paths), <code>integrate_pbe_with_zones</code> integrator with bit-exact 1-zone reduction to legacy <code>PBESolver</code> , <code>consistency_check_with_volume_avg</code> , 31 library tests + 2 CLI smoke tests, <code>dpsim cfd-zones</code> subcommand

Canonical-term additions

In addition to the canonical terms listed above, the following terms are now authoritative in user-facing documentation:

Preferred term	Avoid
<code>epsilon_avg_W_per_kg</code> (volume-average ϵ , drives coalescence)	"average epsilon" without the per-kernel role
<code>epsilon_breakage_weighted_W_per_kg</code> (breakage-frequency-weighted ϵ , drives breakage)	"weighted epsilon", "biased epsilon"
Zone (well-mixed compartment in the CFD-PBE coupling)	"compartment" used loosely; reserve "compartment" for CFD-mesh subdivisions before zonal aggregation
Convective exchange (one-way well-mixed droplet transport between zones)	"mixing flow", "circulation"
Slot exit (rotor-stator-specific high-shear region just outside each stator hole)	"slot region", "stator zone"
<code>qualitative_trend</code> evidence for unvalidated CFD predictions	implying CFD predictions are quantitatively reliable before PIV validation

Scientific and operational boundaries (extension)

The 2026-04-25 audit listed eight interpretation gates for lifecycle results. The CFD-PBE coupling adds three more, applicable when M1 is run via the zonal path:

- PIV validation status of the underlying CFD field (no PIV → predictions inherit `qualitative_trend`).

- `volume_balance_relative_error` < 1e-3 (mass conservation across breakage / coalescence / exchange).
- Consistency check vs the legacy $Po \cdot N^3 \cdot D^5 / V_{\text{tank}}$ empirical estimate (default 30 % relative tolerance per Scientific Advisor guidance, 2026-05-01).

Documentation-update gating (extension)

In addition to the four gates listed above, future updates that touch the CFD-PBE coupling should be gated by:

- 1 Re-run of `tests/test_cfd_zonal_pbe.py` (33 tests, ~32 s wall time).
- 2 Cross-check of `cad/cfd/zones_schema.md` against the Pydantic models

in `src/dpsim/cfd/zonal_pbe.py` — the schema doc is the contract; the models are the implementation; drift between them is a documentation bug.

- 1 PDF rebuild of `appendix_K_cfd_pbe_zonal_coupling.md` alongside the first-edition manual.

Remaining work (Phase C and beyond)

The audit's "remaining documentation risks" section noted that the broader historical `docs/` tree contains archival planning files. As of 2026-05-01, the CFD-PBE coupling has the following items not yet documented in user-facing material because the upstream pipeline has not been executed end-to-end:

- OpenFOAM dictionary templates for Stirrer A and Stirrer B cases (Phase 1-3).
- `extract_epsilon.py` implementation (Phase 4).
- `openfoam_io.py` helper functions.
- A worked end-to-end example with real (rather than synthetic-fixture) `zones.json`.
- PIV validation campaign results.

These will be folded in either via a new audit (`DPSIM_UNIFIED_DOCUMENTATION_AUDIT_<date>.md`) or as a follow-up addendum to this document, depending on whether the broader documentation set has shifted by then.